

Arnav Deepak Tiwari

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Professional Summary

Product Engineer & Automation Specialist with hands-on experience building AI-powered automation systems, data-backed operational platforms, and user-centric product solutions. Delivered measurable business impact at Hitachi India (54% reduction in hazardous incidents, 80% faster document automation via n8n workflows) and Payatu Security (50% faster page loads, 40% backend improvement). Specializes in workflow automation, multi-agent AI systems, and cross-functional product execution in high-growth startup environments.

Technical Skills

Product & Strategy: Product Analytics, User Behavior Analysis, Funnel Analysis, Retention Optimization, Workflow Optimization, Operational Efficiency, Cross-Functional Collaboration, Experimentation-Driven Development, Data-Driven Decision Making

AI & Automation: OpenAI APIs, Google Gemini API, n8n Workflow Automation, LangChain, Prompt Engineering, Multi-Agent Orchestration, RAG, AI Workflow Pipelines

Data & Analytics: SQL, PostgreSQL, MongoDB, MySQL, SQLAlchemy, Drizzle ORM, Python, Data Analysis, Business Metrics

Languages & Engineering: C, C++, Java, JavaScript, TypeScript, React.js, Next.js, Redux, Node.js, Express.js, FastAPI, Django, REST APIs, GraphQL, JWT, OAuth 2.0, WebSockets

DevOps & Cloud: Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux, Bash, Vercel (Cloud Deployment)

Professional Experience

Technical Product & Automation Intern

July 2025 – Present

Hitachi India Pvt Ltd. | Nagpur, Maharashtra

- Drove a **54% reduction in hazardous incidents** and **33% decrease in motor failures** by building a real-time SLD monitoring product that digitalized manual equipment analysis workflows for steel plant operations – providing operations teams with live visibility into critical systems and business metrics.
- Eliminated fragmented manual workflows across 5 HICADS systems (DPS, RCTMS, PTMS, TTMS, ICQIS) by building unified React + Django digital interfaces, reducing operator time consumption by **57%** and improving cross-functional work efficiency by **43%**.
- Built an n8n-powered document automation pipeline reducing turnaround from **2 weeks to 2 days (80% faster)** and scaling weekly delivery to **7–10 project documents** – resolving a critical operational bottleneck and freeing team bandwidth for higher-value work.

Full Stack Developer Intern

May 2025 – July 2025

Payatu Security Consulting Pvt. Ltd. | Pune, Maharashtra

[Certificate](#)

- Improved user experience by optimizing Next.js + SCSS frontend interfaces, reducing page load time from **3s to 1.5s (50% faster)** and enhancing cross-platform UI maintainability.
- Engineered 15 REST API endpoints and optimized 8 existing APIs via indexing, query optimization, and caching on a Node.js + Express.js backend, improving response time by **40%** in collaboration with cross-functional teams.
- Built a real-time event logging product handling **1,000+ log events/minute** for a 4-role RBAC (Admin, System Engineer, Technician, Employee) WiFi admin dashboard – enabling operations teams to monitor system activity and platform health across **1,000+ concurrent users**.

Projects

Autopilot – AI-Powered Project Intelligence Platform

[Link](#) | [GitHub](#)

Next.js, TypeScript, React.js, Node.js, Express.js, PostgreSQL, MongoDB, Google Gemini, Groq SDK, OAuth 2.0, Vercel

- Architected a multi-agent AI planning platform with 4 specialized parallel agents (Gemini + Groq SDK), reducing project plan generation from **10–15 minutes to under 3 minutes (83% faster)** vs. manual coordination workflows.
- Designed an 18-endpoint RESTful API layer orchestrating agent coordination and returning structured execution plans with confidence scores, cost estimations, and task breakdowns – enabling data-driven, cross-functional project decisions at scale.
- Delivered OAuth 2.0 (Google + GitHub) auth, PostgreSQL integration, multi-format export (PDF, DOCX, JSON), i18n, and WCAG-compliant dark-mode UI deployed on Vercel – a scalable, user-centric automation product.

SkillBridge – AI Adaptive Learning & Skill Gap Analytics Engine

[Link](#) | [GitHub](#)

React, FastAPI, Python, FAISS, spaCy, Sentence-Transformers, NetworkX, Docker, SQLAlchemy

- Built an AI skill gap analytics engine addressing learner drop-off, using a **3-layer matching pipeline** (exact match, skill family, cosine similarity via FAISS) trained on **2,484 resumes and 2,277 JDs** – achieving **skill extraction F1 > 0.82** and **gap detection recall > 85%**.
- Designed an original **Weighted Graph Traversal (WGT) algorithm** with Kahn's topological sort for prerequisite-aware learning pathway generation across a **71-skill taxonomy and 58-module** curated course catalog.
- Delivered personalized learning roadmaps in **under 100 seconds** with React DAG visualization (react-flow), real-time 6-stage processing pipeline, LLM reasoning traces, and Docker deployment.

AI-Powered Content Production Platform

[Link](#) | [GitHub](#)

Next.js, React.js, Node.js, TypeScript, Clerk, Drizzle ORM, PostgreSQL, Tailwind CSS, Razorpay, Google Gemini API

- Engineered a scalable LLM-powered content production platform with **18 dynamic templates** spanning blog posts, ad copy, social media, and SEO content – reducing content creation time by **70%** vs. manual workflows.
- Built Node.js + PostgreSQL backend with Drizzle ORM featuring schema-level data integrity, query optimization, and full CRUD support for AI generation history and user analytics data.
- Implemented Razorpay subscription billing with webhook-based automated plan management and Clerk authentication with usage-based credit tracking – a complete consumer SaaS product with end-to-end revenue model.

Achievements

- Rated **5★ (rating 2029)** on CodeChef, ranking among top competitive programmers globally.
- 1st Runner Up – ArtPark Codeforge Hackathon** (Unstop) – competed against teams across India in a software development challenge.
- Achieved **97.6 percentile (AIR 22,007)** in JEE Mains 2022 among 1.1M+ candidates.

Education

Indian Institute of Information Technology Nagpur (IIIT Nagpur)

November 2022 – July 2026

BTech in Computer Science Engineering | **CGPA: 7.34/10**